

Visualizing Program Execution in Virtual Reality

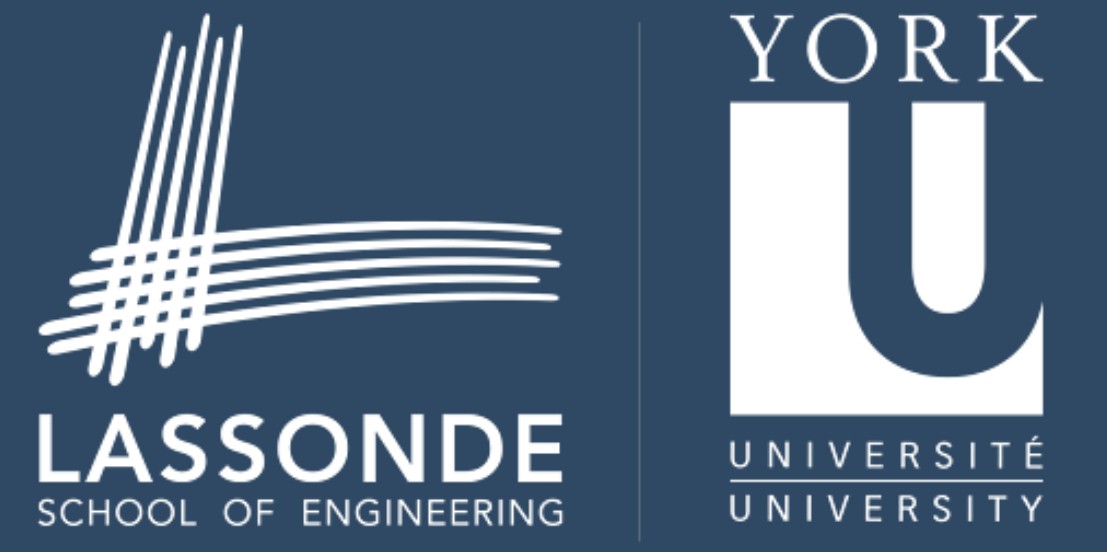
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Introduction

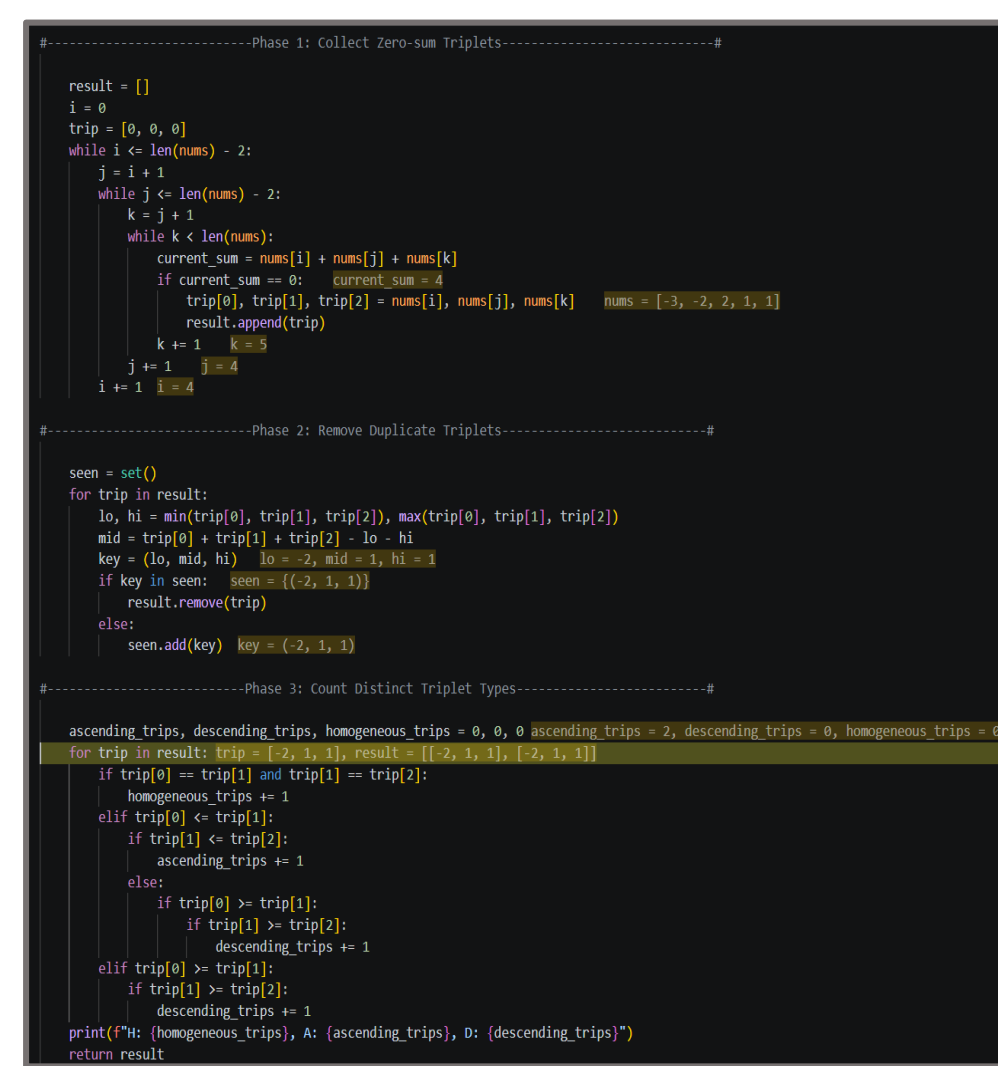


Better understanding
of program execution?

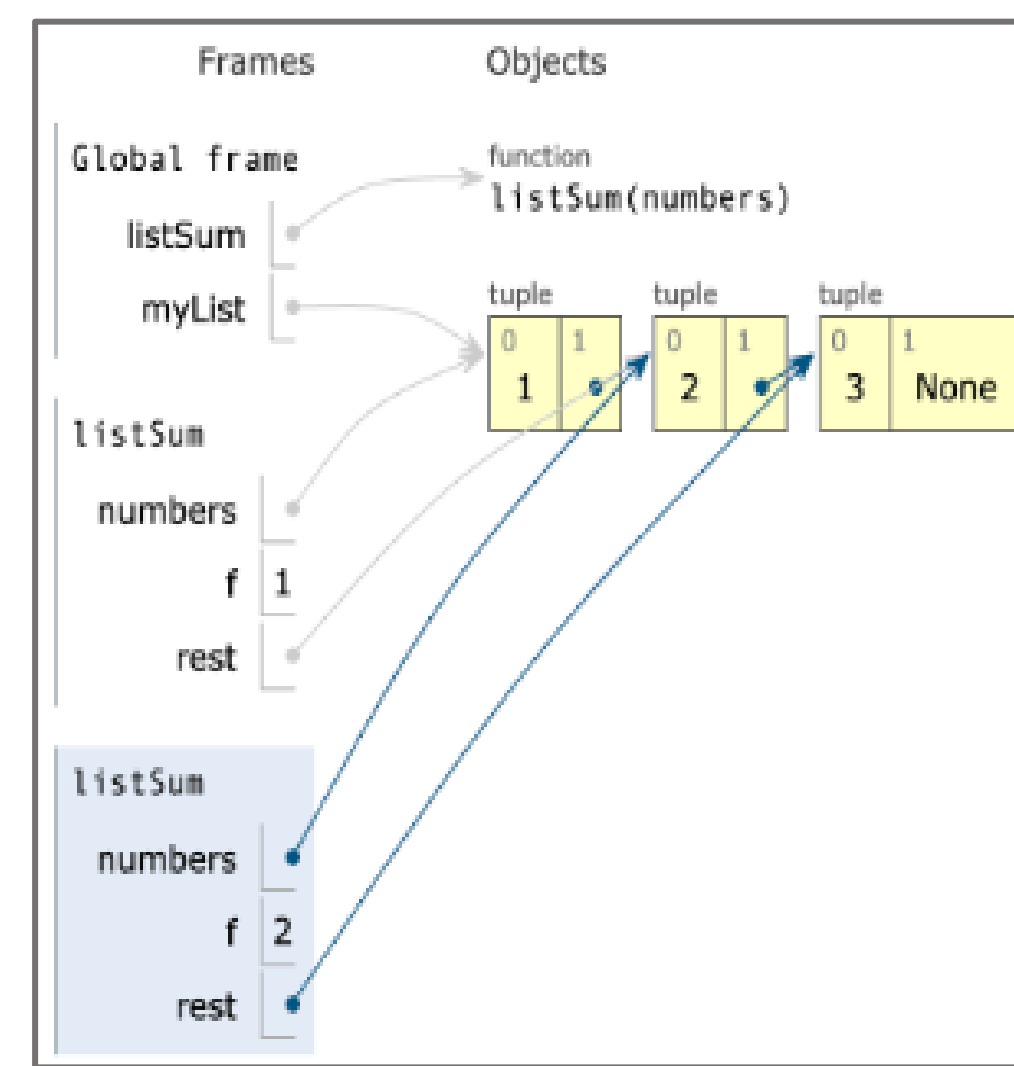
Motivation/Research Question:

- Debugging requires careful reasoning about program execution, changes in program state, and control flow.
- Traditional debuggers expose this information, but can be challenging to use, especially for novices.
- VR offers new ways to spatially represent and interact with program execution.
- **Can VR improve learners' understanding of program execution?**

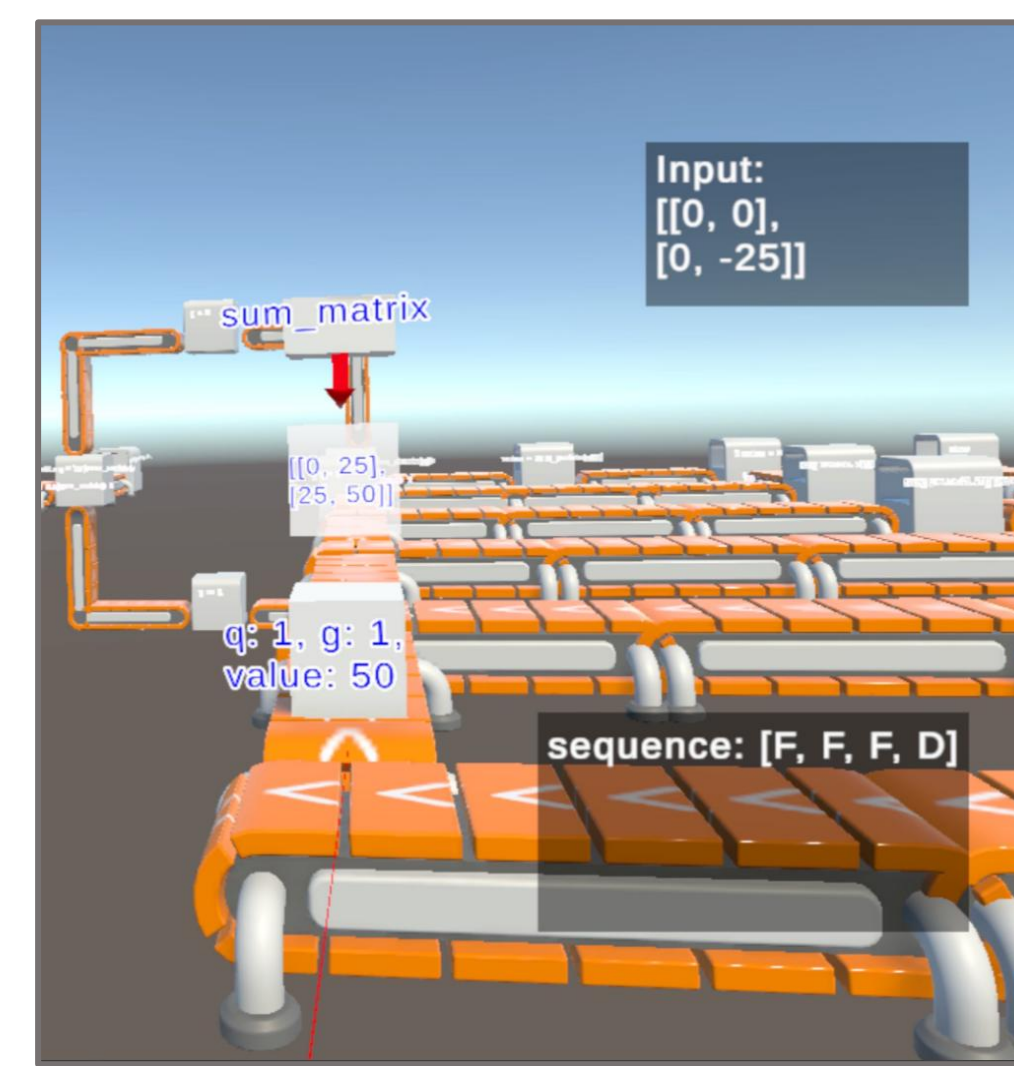
To investigate this question, we built a VR-based debugger and developed a protocol to compare it with other debugging tools, such as VS Code (traditional) and Python Tutor, across three common bug categories: loops, conditionals, and references/initializations.



(a) Traditional Debugger



(b) Python Tutor



(c) VR Debugger

Figure 1. Comparison of the three debugging environments used in the study.

Methods and Materials

Evaluation Materials:

- **3 debuggers:** Traditional (VS Code; D1), Python Tutor (D2), and the VR Debugger (D3).
- **3 tasks:** T1, T2, and T3.
- Each task has **3 versions** (bug categories): Loop (V1), Conditional (V2), Reference/Initialization (V3).
- VR Debugger developed in Unity for Meta Quest 3.

Experimental Design:

- **Planned sample size of n = 100;** all participants must have basic Python knowledge.
- Each participant completes three unique tasks, using a different task version and debugger for each task.
- Debugger and task-version order are counterbalanced to reduce learning, fatigue, and difficulty effects.
- Performance is assessed using debugging accuracy and completion time, followed by a post-study survey of participant experience.

T1	T2	T3
V1D1	V2D2	V3D3
V1D2	V2D3	V3D1
V2D3	V1D1	V3D2
V2D1	V3D2	V1D3
V3D2	V2D3	V1D1
V3D3	V1D1	V2D2
...	...	...

Table 1. Possible trial permutations.

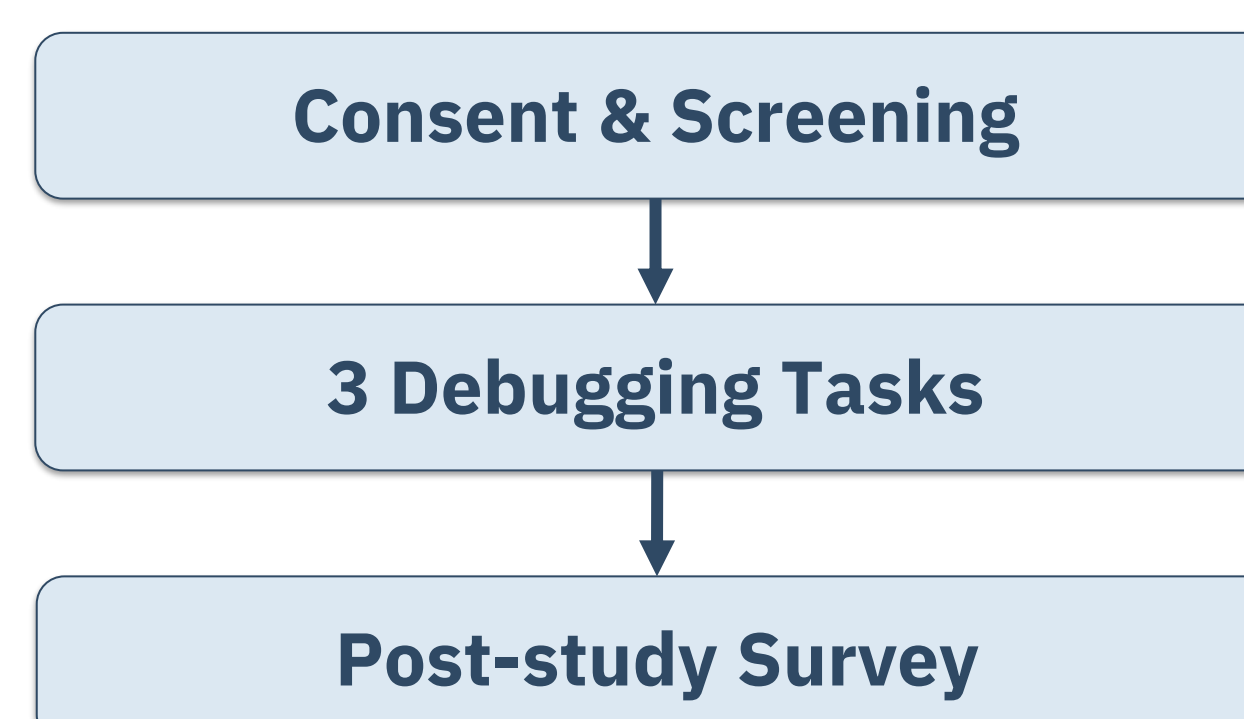


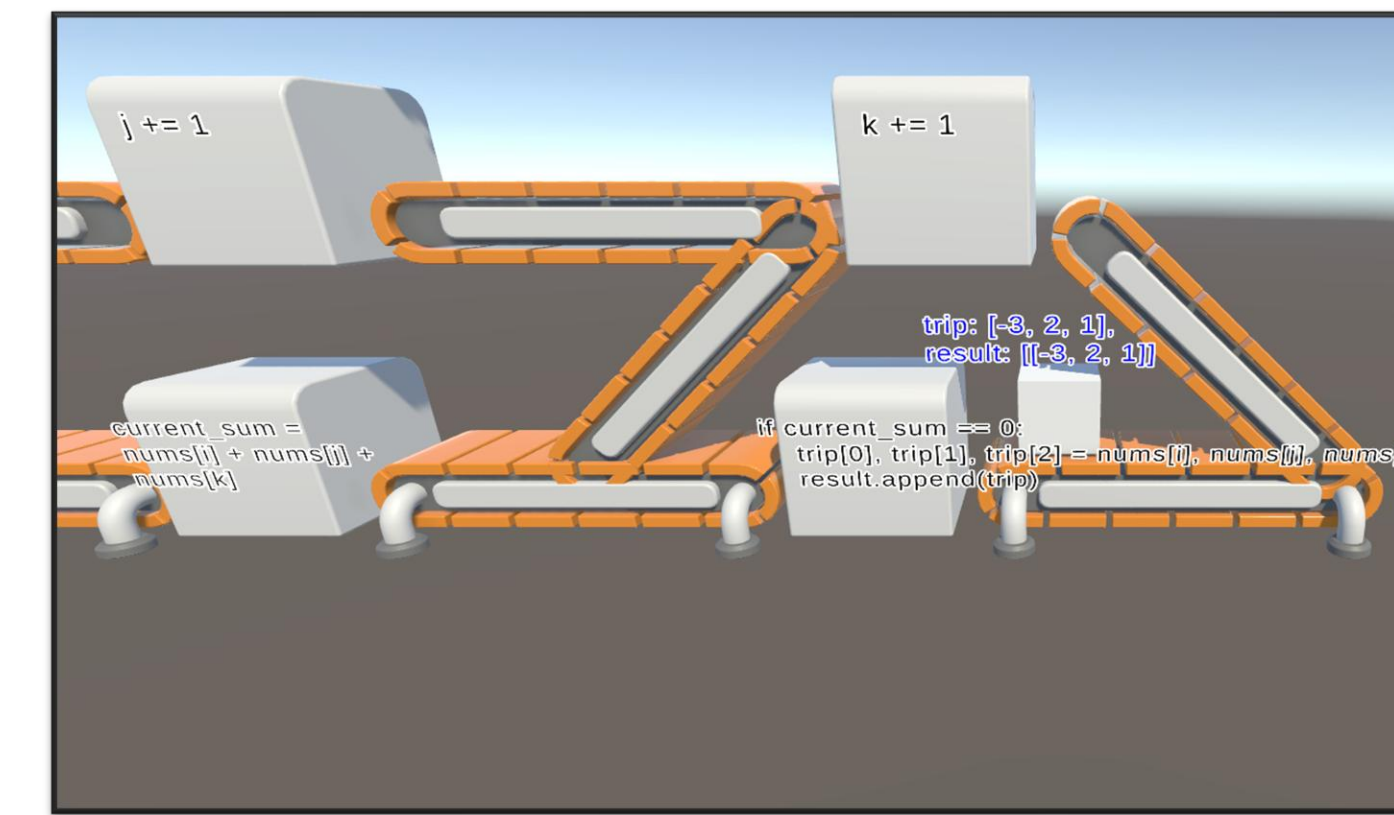
Figure 2. Study procedure.

Results

VR Debugger Development

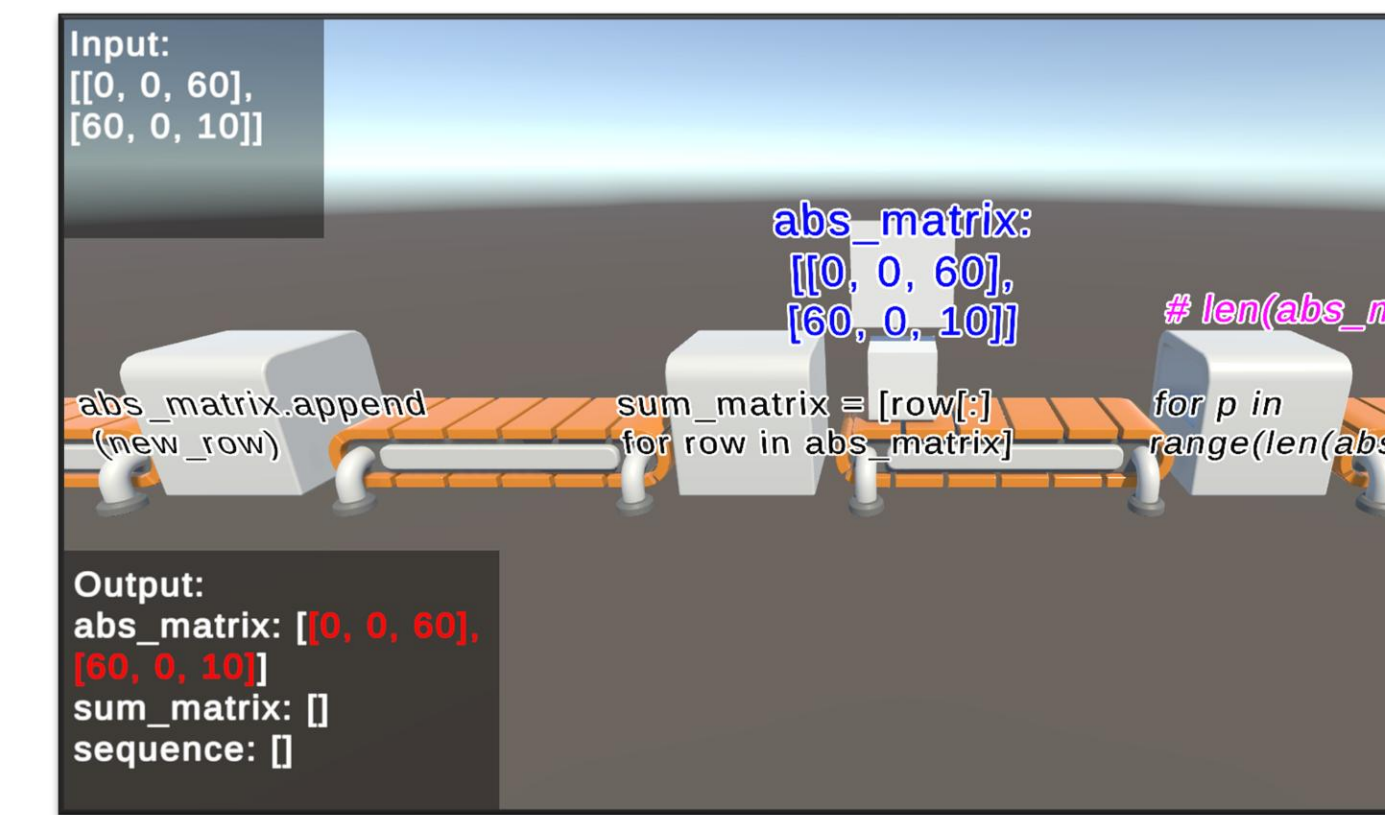
Iteration 1:

Simple, has basic variable tracking.



Iteration 2:

Added input/output dashboard.



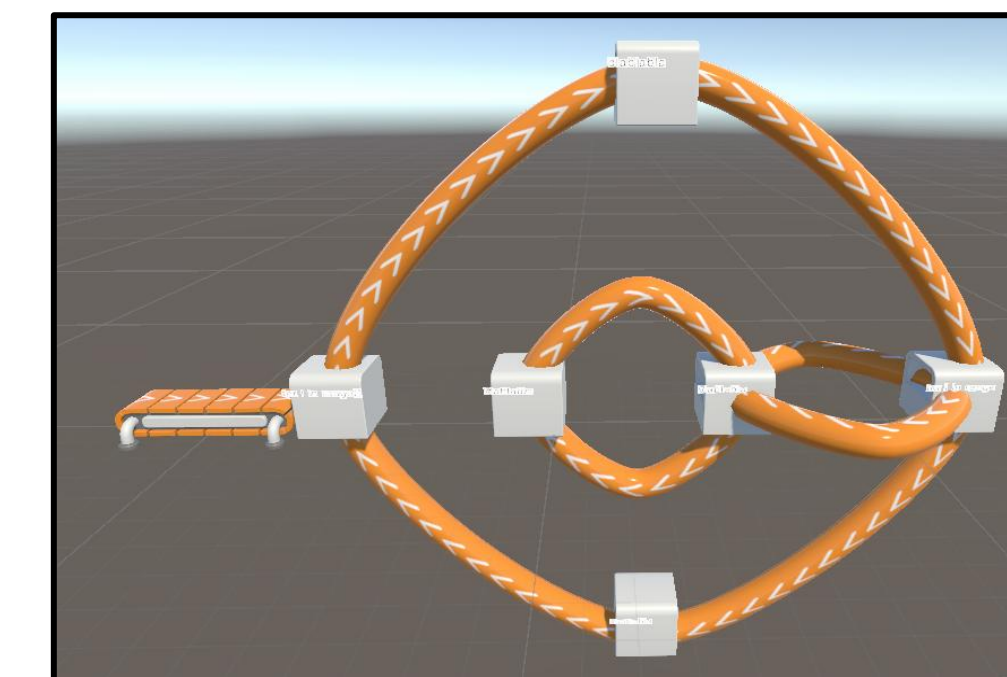
Iteration 3:

Introduced full-code tracking, user controls, and belt direction arrows. Expanded variable tracking and added memory updates.



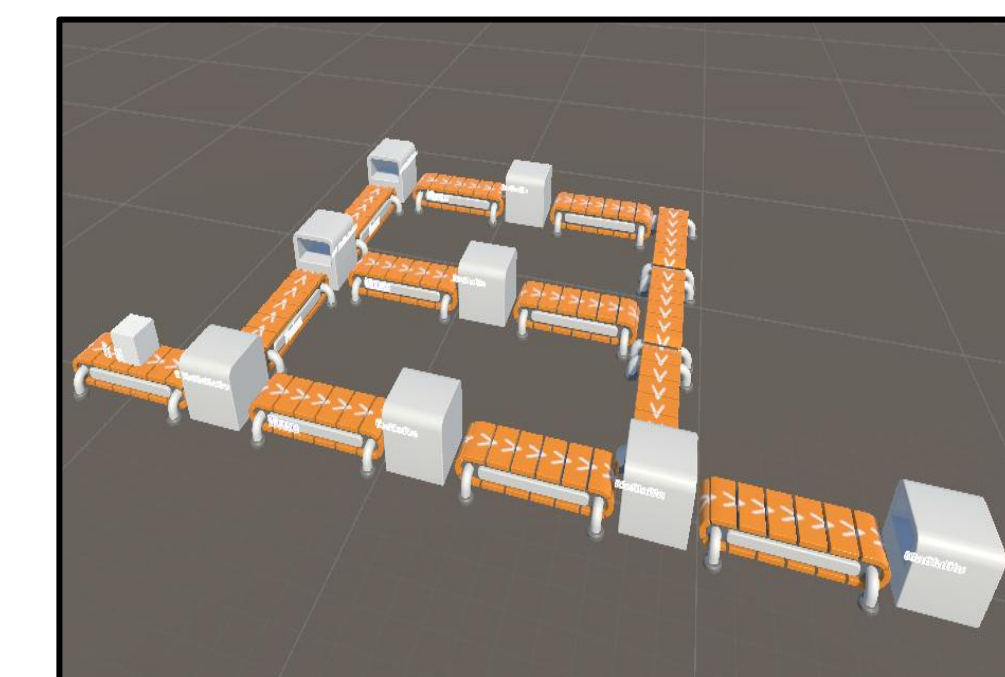
Figure 3. Evolution of the VR debugger across three design iterations.

Final Representations of Bug Categories



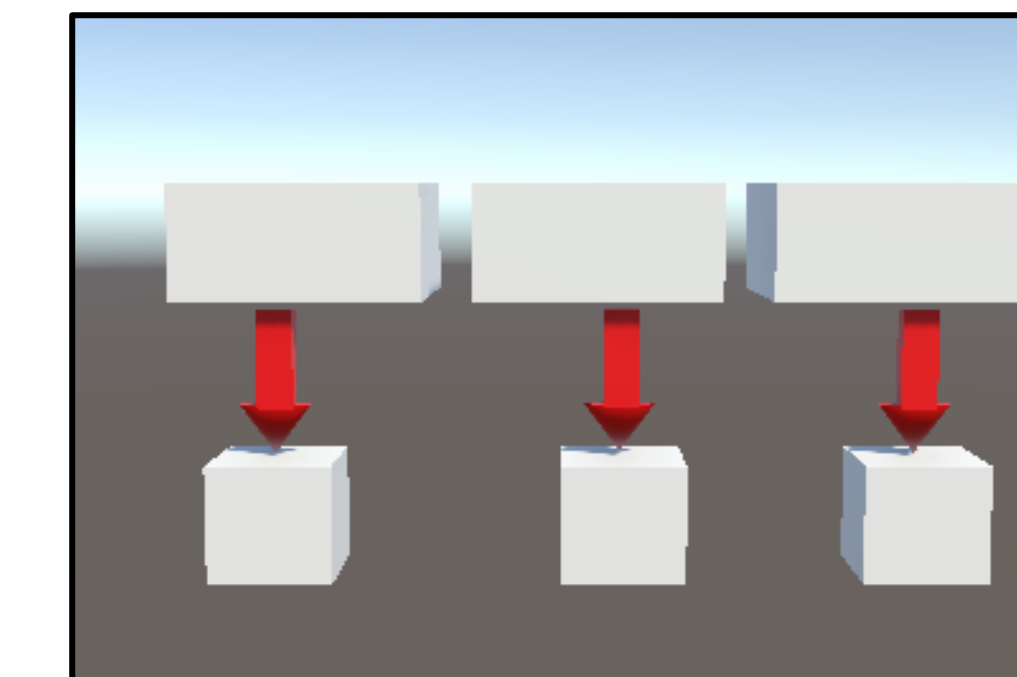
(a) Loop:

Closed-loop tracks represent repetition and cycle logic.



(b) Conditional:

Conveyor junctions represent branching decisions.



(c) Reference/Initialization:

Directional arrows represent memory allocation and object references.

Figure 4. Spatial structures used in the VR debugger to represent the three bug categories.

VR System Capabilities

- Tracks program execution
- Displays changing variables and states
- Visually represents all 3 bug categories
- Provides user controls for stepping through execution

Evaluation Framework Developed

- 3 tasks (9 total versions)
- 3 debugger implementations
- Counterbalanced design
- Measures of debugging performance, and perceived usability/experience

A complete experimental protocol was developed and is reported in the **Methods and Materials** section

Conclusion

Project Status

- All debugging conditions (Traditional, Python Tutor, VR) developed.
- Protocol for evaluating the approach against conventional debugging tools established.
- REB application approved.
- Internal pilot testing underway.

Future Work

- Preliminary feedback on the VR debugger has been positive and is consistent with prior work on embodied cognition^[1] and VR-based learning^[2].
- Formal trials expected to begin in Fall 2026.



Figure 5. York student pilot testing the VR Debugger.

References

- [1] L. W. Barsalou, "Grounded Cognition," Annual Review of Psychology, vol. 59, no. 1, pp. 617–645, Jan. 2008, doi: 10.1146/annurev.psych.59.103006.093639.
- [2] M. Conrad, D. Kablitz, and S. Schumann, "Learning effectiveness of immersive virtual reality in education and training: A systematic review of findings," Computers & Education: X Reality, vol. 4, p. 100053, Jan. 2024, doi: 10.1016/j.cexr.2024.100053.

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